CRUDE ASSAY REPORT
Report Number: IDL-CA-0028-2024
SAMPLE DESCRIPTION: Qatar Marine Crude Oil - Qatar Marine Loading dated 25-02-2024

Bureau Veritas Commodities Division
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LABORATORY REPORT NO. CA-0028-2024

Date: April 8, 2024

Company details	:	Qatar Petroleum for the Sale of Petroleum Products Company Limited (QPSPP) P. O. Box 24183, Doha, Qatar
For the Attention of	:	Mr. Raphael Chike Philips
Sample(s) received from	:	Qatar Energy – Halul Island, Qatar
Description(s) on Label(s)	:	Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024
Seals on Sample(s)	:	43027,43052,43730,43760 & 43932
Sample(s) received on	:	06-03-2024

The above sample(s) was/were examined as detailed below and the following results obtained

Please refer attached sheet for analytical results.

REPORTED BY

Rejeesh Reghu

Sr. Chemist

AUTHORIZED SIGNATURE

Biju GEORGE

Assay Project Manager

LABORATORY REPORT NO. CA-0028-2024

Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024

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CRUDE OIL - DETAILED ASSAY ALL CUTS OVERVIEW
Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024

TEST	METHOD	UNIT	WHOLE CRUDE	LPG	FRACTIONS												
					NAPHTHA			KEROSENE		DISTILLATE FUELS (GO)			VGO			ATM. RESIDUE	VACUUM RESIDUE
Initial BP					**C5**	**50**	**100**	**150**	**200**	**250**	**300**	**350**	**400**	**450**	**500**	**400 + Res.**	**550 + Res.**
Final BP		**°C**			**50**	**100**	**150**	**200**	**250**	**300**	**350**	**400**	**450**	**500**	**550**		
Distillation																	
Yield on Crude	ASTM D 2892 / D5236	% Wt.		1.30	2.08	6.10	7.51	8.37	8.42	8.68	8.65	8.06	6.12	6.27	6.54	40.81	21.89
Yield on Crude		% Vol		1.94	2.83	7.56	8.67	9.23	8.94	8.89	8.53	7.69	5.76	5.80	5.94	35.72	18.22
Density @ 15°C	ASTM D5002 / D4052 / IP 365	kg/L	0.8694		0.6376	0.7018	0.7536	0.7888	0.8188	0.8489	0.8815	0.9122	0.9239	0.9394	0.9565	0.9933	1.0442
Specific Gravity @ 60/60° F	Conversion		0.8699			0.7020	0.7538	0.7892	0.8192	0.8493	0.8820	0.9127	0.9244	0.9399	0.9571	0.9939	1.0449
API Gravity @ 60° F	Calculated	° API	31.2			70.1	57.5	47.8	41.2	35.1	28.9	23.5	21.6	19.1	16.3	10.9	3.9
Sulphur	ASTM D4294	% Wt.	2.10		0.0442	0.0510	0.0553	0.138	0.346	1.07	1.96	2.40	2.52	2.90	3.21	3.90	4.78
Total Acid Number	ASTM D664 / D974	mgKOH/g	0.21					0.02	0.04	0.05	0.05	0.06	0.06	0.07	0.08		
Pour Point	ASTM D97 / D5853	°F	0.0					< - 27.0		5.0	37.0	64.0					
Kinematic Viscosity @ 15°C	ASTM D445	cSt	14.20														
Kinematic Viscosity @ 40°C	ASTM D445	cSt	7.260														
Kinematic Viscosity @ 122°F (50.0°C)	ASTM D445	cSt							1.242	2.247	4.432	9.518	28.86	48.43	125.9		
Kinematic Viscosity @ 210°F (98.9°C)	ASTM D445	cSt							0.668	1.081	1.940	3.374	7.204	10.17	18.88	99.9	3587
Kinematic Viscosity @ 275°F (135°C)	ASTM D445	cSt														27.10	367.0
Kinematic Viscosity @ 350°F (176.7°C)	ASTM D445	cSt														9.914	65.99
Benzene	IP 601	%Wt.	0.97														
Benzene	ASTM D6730	% Vol			0.099	2.299	0.261										
Cyclo C6	ASTM D6730	% Vol			<0.1	3.366	0.645										
Methyl-cyclopentane	ASTM D6730	% Vol			0.091	5.487	0.421										
C6 Paraffins	ASTM D6730	% Vol			7.497	39.303	0.896										
Reid Vapour Pressure	ASTM D5191	psi	6.90		17.4	5.30	0.95										
Hydrocarbon Types																	
Total Paraffins	ASTM D6730	% Vol				77.473	64.535	46.463									
Olefins		% Vol				<0.1	0.451	3.968									
Naphthenes		% Vol				17.350	21.599	29.491									
Aromatics		% Vol				5.177	13.415	20.078									
Total Paraffins	ASTM D6730	% Wt.				74.473	61.039	43.751									
Olefins		% Wt.				<0.1	0.439	3.776									
Naphthenes		% Wt.				19.007	22.898	30.085									
Aromatics		% Wt.				6.520	15.624	22.388									
Smoke Point	ASTM D1322	mm						26	24								
Freezing Point	ASTM D2386	°F							-46.0								
Cetane Index	ASTM D976								47.4	50.0	47.9						
Metal																	
Nickel	IP 501	ppm wt.	12													29	55
Vanadium		ppm wt.	31													76	141
Iron		ppm wt.	11													27	50
Potassium		ppm wt.	2														
Magnesium		ppm wt.	<1														
Sodium		ppm wt.	50														
Calcium		ppm wt.	3														
Copper		ppm wt.	<1														
Chromium		ppm wt.	<1														
Zinc		ppm wt.	<1														
Manganese		ppm wt.	<1														
Aluminium		ppm wt.	<1														
Cadmium		ppm wt.	<1														
Lead		ppm wt.	<1														
Mercury		ppm wt.	<1														
Mercaptan Sulphur	UOP 163	ppm wt.	48		336	225	166	132	44								
Aniline Point	ASTM D611	° C						53.0	58.0	60.0	63.0	67.0	72.0	78.0			
Distillation	ASTM D86				See page 13	See page 13	See page 13	See Page 13	See Page 13	See Page 13	See Page 13	See Page 13					
UOP K Factor	UOP 375	calc.										11.5	11.7				
Light Ends																	
Methane	IP 344	% Wt.	0.000														
Ethane			0.000														
Propane			0.098														
i-Butane			0.128														
n- Butane			0.539														
i-Pentane			0.573														
n- Pentane			0.865														
Total Nitrogen	ASTM D5762	ppm Wt										504		1025		2582	3487
Hydrogen Sulfide	UOP 163	ppm wt.	<1														
Salt Content	ASTM D3230	PTB	45.2														
Salt Content	ASTM D3230	mg/L	148.4														
Toluene Equivalent	Exxon Ams 79-004	%	23.0														
Xylene Equivalent	BP 230	%	22.5														
True Vapor Pressure	ASTM D2889	psi	8.17														
Auto Ignition Temperature		°C	237														
Ash Content	ASTM D482	% Wt.	0.008														
Flash Point	ASTM D93	°C	< - 5														
Upper Flammable Limit		%v/v	2.4														
Lower Flammable Limit		%v/v	0.3														
Wax Content	UOP 46	% Wt.	<5														
Conradson Carbon	ASTM D189	% Wt.									0.02		0.05		1.04	13.4	24.7

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TRUE BOILING POINT DISTILLATION DATA

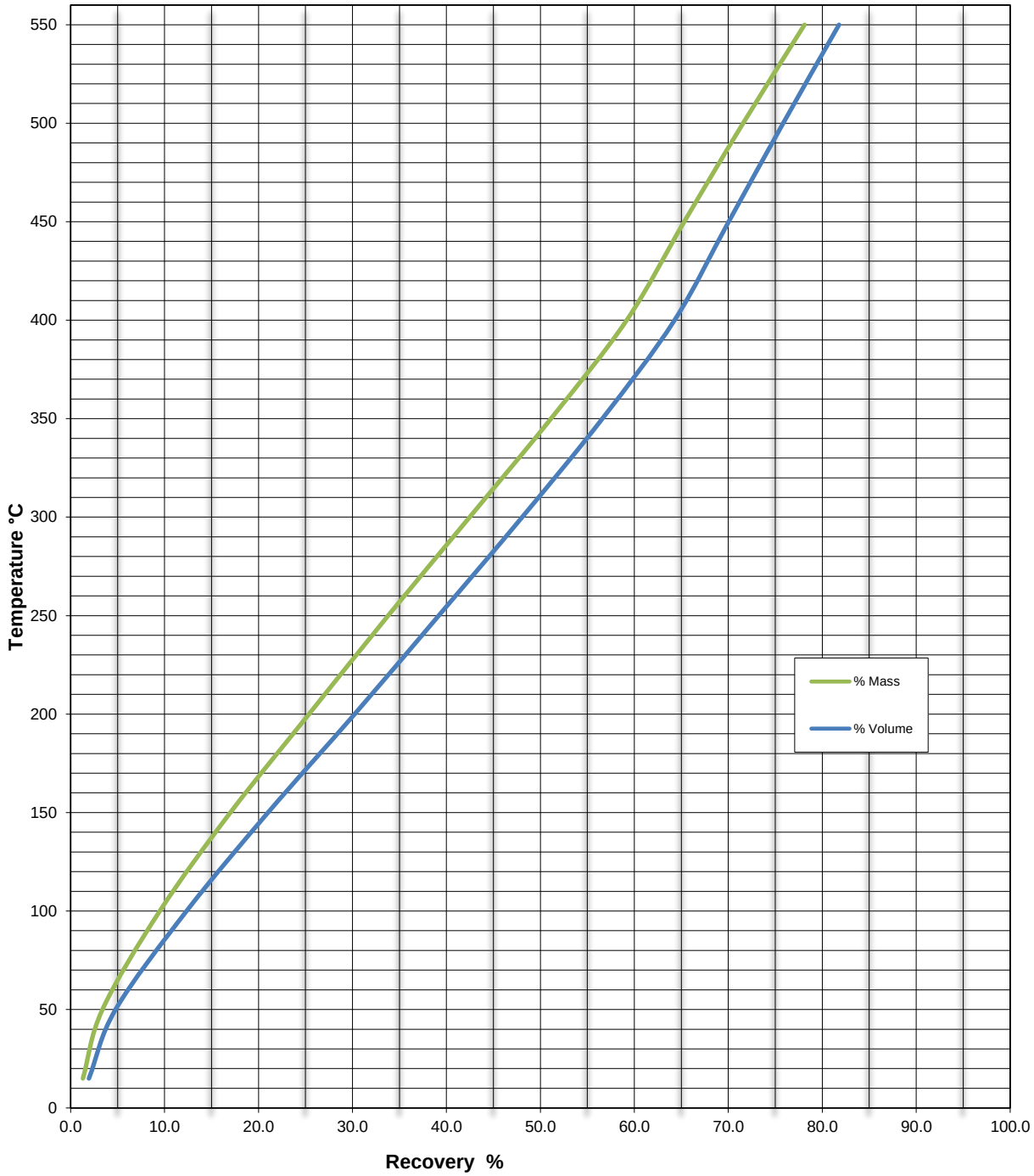
(ASTM D2892 & ASTM D5236)

Sample Descriptions / Label : **Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024**

Sl. No.	Vapour Temperature °C	% Mass	Cumulative % Mass	% Volume	Cumulative % Volume
1	Gas	1.30	1.30	1.94	1.94
2	15 - 50	2.08	3.38	2.83	4.78
3	50 - 100	6.10	9.48	7.56	12.34
4	100 - 150	7.51	17.00	8.67	21.00
5	150 - 200	8.37	25.37	9.23	30.23
6	200 - 250	8.42	33.79	8.94	39.17
7	250 - 300	8.68	42.47	8.89	48.06
8	300 - 350	8.65	51.12	8.53	56.59
9	350 - 400	8.06	59.19	7.69	64.28
10	400 - 450	6.12	65.31	5.76	70.04
11	450 - 500	6.27	71.57	5.80	75.84
12	500 - 550	6.54	78.11	5.94	81.78
13	550 + Residue	21.89	100.00	18.22	100.00

TRUE BOILING POINT DISTILLATION CURVE

LABORATORY REPORT NO. CA-0028-2024



LABORATORY REPORT NO. CA-0028-2024

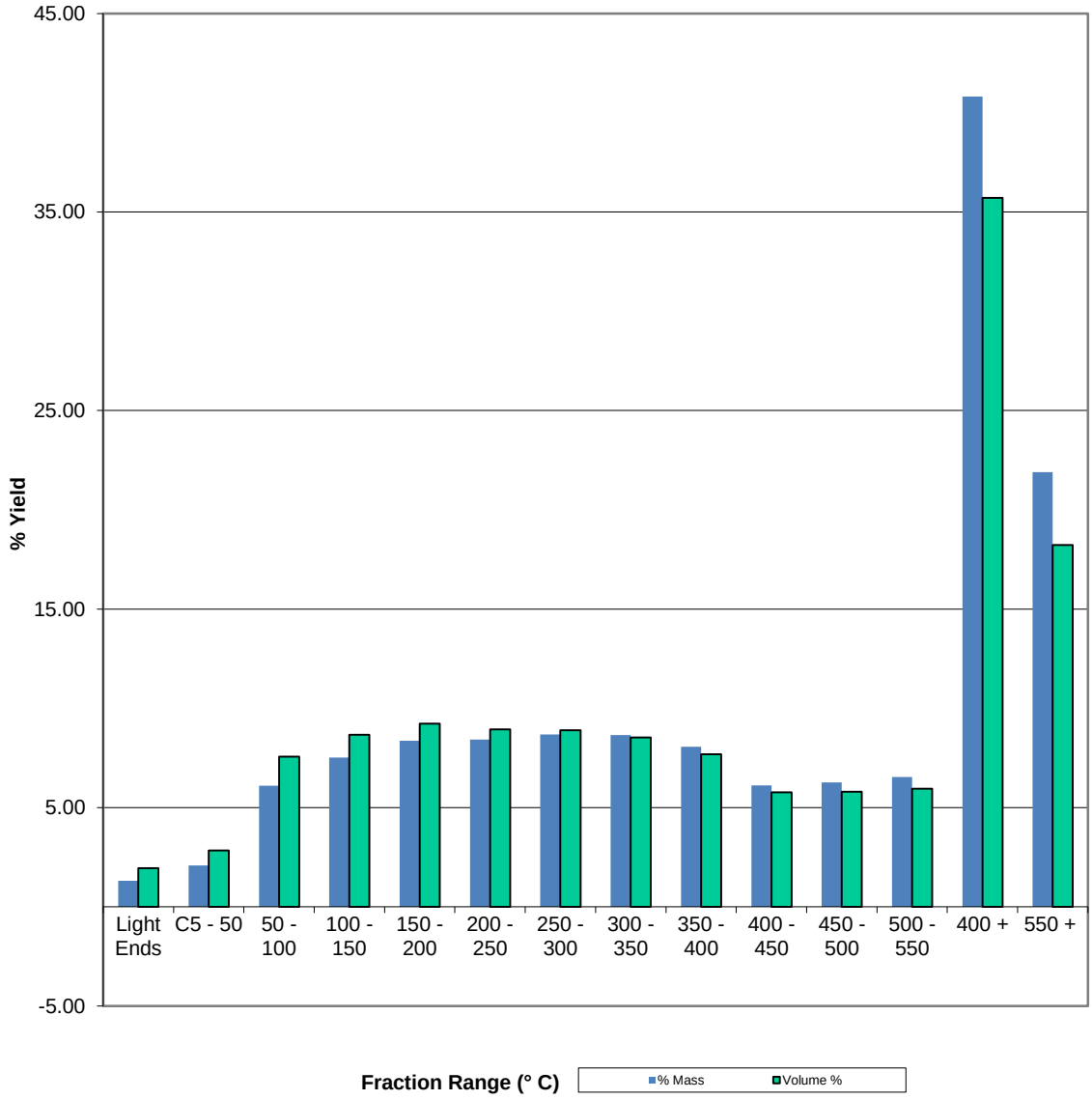
SUMMARY OF PRODUCT / RESIDUE CUT POINTS AND YIELDS

Sample Descriptions / Label: **Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024**

Yield %			
Product	Cut Points (° C)	Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024	
		% Mass	Volume %
GAS	Light Ends	1.30	1.94
NAPHTHA	C5 - 50	2.08	2.83
	50 - 100	6.10	7.56
	100 - 150	7.51	8.67
KEROSENE	150 - 200	8.37	9.23
	200 - 250	8.42	8.94
GAS OIL	250 - 300	8.68	8.89
	300 - 350	8.65	8.53
	350 - 400	8.06	7.69
VACUUM GAS OIL	400 - 450	6.12	5.76
	450 - 500	6.27	5.80
	500 - 550	6.54	5.94
ATM. RESIDUE	400 +	40.81	35.72
VACUUM RESIDUE	550 +	21.89	18.22

SUMMARY OF TRUE BOILING POINT DISTILLATION WIDE CUTS

LABORATORY REPORT NO: IDL-CA-0016-2024



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Whole Crude Properties

Sample Descriptions / Label : **Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024**

TEST	METHOD	UNIT	RESULT
Density @ 15°C	ASTM D 5002	kg/L	0.8694
Specific Gravity @ 60/60° F	Conversion		0.8699
API Gravity @ 60° F	Calculated	° API	31.2
Sulphur	ASTM D4294	% Wt.	2.10
Total Acid Number	ASTM D664	mgKOH/g	0.21
Pour Point	ASTM D5853	°F	0
Kinematic Viscosity @ 15°C	ASTM D445	cSt	14.20
Kinematic Viscosity @ 40°C	ASTM D445	cSt	7.260
Reid Vapour Pressure	ASTM D5191	psi	6.90
Metal			-
Nickel	ICP	ppm wt.	12
Vanadium		ppm wt.	31
Iron		ppm wt.	11
Potassium		ppm wt.	2
Magnesium		ppm wt.	<1
Sodium		ppm wt.	50
Calcium		ppm wt.	3
Copper		ppm wt.	<1
Chromium		ppm wt.	<1
Zinc		ppm wt.	<1
Manganese		ppm wt.	<1
Aluminium		ppm wt.	<1
Cadmium		ppm wt.	<1
Lead		ppm wt.	<1
Mercury		ppm wt.	<1
Mercaptan Sulphur	UOP 163	ppm wt.	48.0
Light Ends			-
Methane	IP 601	% Wt.	0.000
Ethane			0.000
Propane			0.098
i-Butane			0.128
n- Butane			0.539
i-Pentane			0.573
n- Pentane			0.865
Hydrogen Sulfide	UOP 163	ppm wt.	<1
Salt Content	ASTM D3230	PTB	45.2
Salt Content	ASTM D3230	mg/L	148.4
Toluene Equivalent	EXXON 79-001	%	23.0
Xylene Equivalent	BP 230	%	22.5
True Vapor Pressure	ASTM D2889	psi	8.17
Auto Ignition Temperature	ASTM E 659	°C	237
Ash Content	ASTM D482	% wt.	0.008
Flash Point	ASTM D93	°C	< - 5
Upper Flammable Limit		%v/v	2.40
Lower Flammable Limit		%v/v	0.25
Benzene	IP 601	% Wt.	0.97
Wax Content	UOP 46	% Wt.	<5

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Naphtha Fractions

Sample Descriptions: **Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024**

TEST	METHOD	UNIT	NAPHTHA FRACTIONS		
Initial BP		**°C**	**C5**	**50**	**100**
Final BP		**°C**	**50**	**100**	**150**
Yield	ASTM D2892 /D5236	% Wt.	2.08	6.10	7.51
Yield		% Vol.	2.83	7.56	8.67
Density @ 15°C	ASTM D4052	kg/L	0.6376	0.7018	0.7536
Specific Gravity @ 60/60° F	Conversion	-		0.7020	0.7538
API Gravity @ 60° F	Calculated	° API		70.1	57.5
Sulphur Content	ASTM D4294	% Wt.	0.0442	0.0510	0.0553
Benzene	ASTM D6730	% Vol.	0.099	2.299	0.261
Cyclo C6	ASTM D6730	% Vol	<0.1	3.366	0.645
Methyl-cyclopentane	ASTM D6730	% Vol	0.091	5.487	0.421
C6 Paraffins	ASTM D6730	% Vol	7.497	39.303	0.896
Reid Vapour Pressure	ASTM D5191	psi	17.4	5.30	0.95
Hydrocarbon Types	.	.			
Total Paraffins	ASTM D6730	% Vol		77.473	64.535
Olefins		% Vol		<0.1	0.451
Naphthenes		% Vol		17.350	21.599
Aromatics		% Vol		5.177	13.415
Total Paraffins	ASTM D6730	% Wt.		74.473	61.039
Olefins		% Wt.		<0.1	0.439
Naphthenes		% Wt.		19.007	22.898
Aromatics		% Wt.		6.520	15.624
Mercaptan Sulphur	UOP 163	ppm	336	225	166
Distillation	ASTM D86	°C	See page 13	See page 13	See page 13

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Kerosene Fractions

Sample Descriptions: **Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024**

TEST	METHOD	UNIT	KEROSENE FRACTIONS	
Initial BP		**°C**	**150**	**200**
Final BP		**°C**	**200**	**250**
Yield	ASTM D2892/D5236	% Wt.	8.37	8.42
Yield		% Vol.	9.23	8.94
Density @ 15°C	ASTM D4052	kg/L	0.7888	0.8188
Specific Gravity @ 60/60° F	Conversion	.	0.7892	0.8192
API Gravity @ 60° F	Calculated	° API	47.8	41.2
Sulphur Content	ASTM D4294	% Wt.	0.138	0.346
Total Acid Number	ASTM D974	mgKOH/g	0.02	0.04
Pour Point	ASTM D97	°F	< - 27.0	
Kinematic Viscosity @ 122°F (50.0°C)	ASTM D445	cSt		1.242
Kinematic Viscosity @ 210°F (98.9°C)	ASTM D445	cSt		0.668
Hydrocarbon Types				
Total Paraffins	ASTM D6730	% Vol	46.463	
Olefins		% Vol	3.968	
Naphthenes		% Vol	29.491	
Aromatics		% Vol	20.078	
Total Paraffins	ASTM D6730	% Wt.	43.751	
Olefins		% Wt.	3.776	
Naphthenes		% Wt.	30.085	
Aromatics		% Wt.	22.388	
Smoke Point	ASTM D1322	mm	26	24
Freezing Point	ASTM D2386	°F		-46.0
Cetane Index	ASTM D976	-		47.4
Mercaptan Sulphur	UOP 163	ppm	132	44
Aniline Point	ASTM D611	°C	53.0	58.0
Distillation	ASTM D86	°C	See Page 13	See Page 13

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Gas Oil Fraction

Sample Descriptions: **Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024**

TEST	METHOD	UNIT	DISTILLATE FUELS (GO)		
Initial BP	.	**°C**	**250**	**300**	**350**
Final BP	.	**°C**	**300**	**350**	**400**
Yield	ASTM D2892 /D5236	% Wt.	8.68	8.65	8.06
Yield		% Vol.	8.89	8.53	7.69
Density @ 15°C	ASTM D4052	kg/L	0.8489	0.8815	0.9122
Specific Gravity @ 60/60° F	Conversion	-	0.8493	0.8820	0.9127
API Gravity @ 60° F	Calculated	° API	35.1	28.9	23.5
Sulphur Content	ASTM D4294	% Wt.	1.07	1.96	2.40
Total Acid Number	ASTM D974	mgKOH/g	0.05	0.05	0.06
Pour Point	ASTM D97	°F	5.0	37.0	64.0
Kinematic Viscosity @ 122°F (50.0°C)	ASTM D445	cSt	2.247	4.432	9.518
Kinematic Viscosity @ 210°F (98.9°C)	ASTM D445	cSt	1.081	1.940	3.374
Cetane Index	ASTM D976	-	50.0	47.9	
Aniline Point	ASTM D611	° C	60.0	63.0	67.0
Distillation	ASTM D86	° C	See Page 13	See Page 13	See Page 13
UOP K Factor	UOP 375	calc.			11.5
Total Nitrogen	ASTM D5762	ppm Wt			504
Conradson Carbon	ASTM D189	% Wt.		0.02	

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Vacuum Gas Oil Fractions

Sample Descriptions: **Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024**

TEST	METHOD	UNIT	VACUUM GAS OIL FRACTIONS		
Initial BP		**°C**	**400**	**450**	**500**
Final BP		**°C**	**450**	**500**	**550**
Yield	ASTM D 2892 /D5236	% Wt.	6.12	6.27	6.54
Yield		% Vol.	5.76	5.80	5.94
Density @ 15°C	ASTM D4052	kg/L	0.9239	0.9394	0.9565
Specific Gravity @ 60/60° F	Conversion		0.9244	0.9399	0.9571
API Gravity @ 60° F	Calculated	° API	21.6	19.1	16.3
Sulphur Content	ASTM D4294	% Wt.	2.52	2.90	3.21
Total Acid Number	ASTM D664	mgKOH/g	0.06	0.07	0.08
Kinematic Viscosity @ 122°F (50.0°C)	ASTM D445	cSt	28.86	48.43	125.9
Kinematic Viscosity @ 210°F (98.9°C)	ASTM D445	cSt	7.204	10.17	18.88
Aniline Point	ASTM D611	° C	72.0	78.0	
UOP K Factor	UOP 375	calc.	11.7		
Total Nitrogen	ASTM D5762	ppm Wt		1025	
Conradson Carbon	ASTM D189	% Wt.	0.05		1.04

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Residues

Sample Descriptions / Label: **Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024**

TEST	METHOD	UNIT	Residue Cuts	
Initial BP		**°C**	**400 +**	**550 +**
Final BP		**°C**		
Yield	ASTM D2892 /D5236	% Wt.	40.81	21.89
Yield		% Vol.	35.72	18.22
Density @ 15°C	IP 365	kg/L	0.9933	1.0442
Specific Gravity @ 60/60° F	Conversion	-	0.9939	1.0449
API Gravity @ 60° F	Calculated	° API	10.9	3.92
Sulphur	ASTM D4294	% Wt.	3.90	4.78
Kinematic Viscosity @ 210°F (98.9°C)	ASTM D445	cSt	99.93	3587
Kinematic Viscosity @ 275°F (135°C)	ASTM D445	cSt	27.10	367.0
Kinematic Viscosity @ 350°F (176.7°C)	ASTM D445	cSt	9.914	65.99
Metal				-
Nickel	IP 501	ppm Wt.	29	55
Vanadium		ppm Wt.	76	141
Iron		ppm Wt.	27	50
Total Nitrogen	ASTM D5762	ppm Wt	2582	3487
Conradson Carbon	ASTM D189	% Wt.	13.4	24.7

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ASTM D 86 - DISTILLATION DATA PER FRACTIONS

Sample Descriptions / Label : Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024

Boiling range	°C	C5	50	100	150	200	250	300	350
		50	100	150	200	250	300	350	400
% Recovery	Unit	Temperature							
Initial boiling point	°C	27.8	49.7	100.4	149.7	200.3	249.5	300.7	350.5
Recovery @ 5%	°C	31.2	64.3	107.1	158.0	210.0	263.0	312.6	348.4
Recovery @ 10%	°C	31.6	70.1	109.5	162.2	212.5	264.1	314.1	358.0
Recovery @ 20%	°C	32.4	72.1	113.0	163.5	216.0	265.3	315.7	360.1
Recovery @ 30%	°C	33.0	74.1	116.8	167.7	217.4	268.4	318.6	361.5
Recovery @ 40%	°C	33.7	76.2	118.0	170.0	219.6	270.2	322.3	362.9
Recovery @ 50%	°C	34.5	78.0	122.3	172.3	220.8	270.0	323.2	365.2
Recovery @ 60%	°C	35.2	80.4	124.2	173.8	223.2	273.9	324.7	370.0
Recovery @ 70%	°C	36.3	83.5	127.7	176.7	226.0	276.6	326.3	371.1
Recovery @ 80%	°C	37.7	87.0	131.9	181.5	228.3	281.0	329.2	373.5
Recovery @ 85%	°C	38.8	88.8	136.6	184.0	232.4	283.4	333.0	376.2
Recovery @ 90%	°C	40.6	92.0	139.5	186.6	237.0	286.8	335.9	382.0
Recovery @ 95%	°C	45.8	95.1	143.5	192.0	240.3	291.7	341.4	390.6
Final Boiling Point	°C	50.3	99.8	150.2	199.5	250.8	300.3	349.6	399.7

Total Recovered	% Vol.	99.3	99.2	99.1	99.0	99.0	98.8	98.5	98.3
Residue	% Vol.	0.3	0.3	0.4	0.5	0.6	0.9	1.2	1.4
Loss	% Vol.	0.6	0.5	0.5	0.5	0.4	0.3	0.3	0.3

LABORATORY REPORT NO. CA-0028-2024

DETAILED HYDROCARBON ANALYSIS

Sample Descriptions / Label : Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024

FRACTION - (50 - 100°C)

Totals by Group Type and Carbon Number (in Mass Percent)

	n-paraffins	I-paraffins	Olefins	Naphthenes	Aromatics	Total
C3						0.000
C4						0.000
C5	5.965	0.992		1.270		8.227
C6	20.425	16.947		9.713	2.918	50.003
C7	13.454	13.376		7.838	3.602	38.270
C8	0.296	3.018		0.186		3.500
C9						0.000
C10						0.000
C11						0.000
Total Unknowns						0.000
Total	**40.140**	**34.333**	**0.000**	**19.007**	**6.520**	**100.000**

Totals by Group Type and Carbon Number (in Volume Percent)

	n-paraffins	I-paraffins	Olefins	Naphthenes	Aromatics	Total
C3						0.000
C4						0.000
C5	6.606	1.110		1.180		8.896
C6	21.455	17.847		8.853	2.299	50.454
C7	13.631	13.541		7.148	2.878	37.198
C8	0.292	2.991		0.169		3.452
C9						0.000
C10						0.000
C11						0.000
Total Unknowns						0.000
Total	**41.984**	**35.489**	**0.000**	**17.350**	**5.177**	**100.000**

LABORATORY REPORT NO. CA-0028-2024

DETAILED HYDROCARBON ANALYSIS

Sample Descriptions / Label : Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024

FRACTION - (100 - 150°C)

Totals by Group Type and Carbon Number (in Mass Percent)

	n-paraffins	I-paraffins	Olefins	Naphthenes	Aromatics	Total
C3						0.000
C4						0.000
C5	0.043					0.043
C6	0.574	0.246		1.112	0.308	2.240
C7	6.260	3.138		5.009	5.008	19.415
C8	15.305	14.525		8.582	8.461	46.873
C9	8.031	10.580	0.429	8.195	1.847	29.082
C10	0.148	2.189	0.010			2.347
C11						0.000
Total Unknowns						0.000
Total	**30.361**	**30.678**	**0.439**	**22.898**	**15.624**	**100.000**

Totals by Group Type and Carbon Number (in Volume Percent)

	n-paraffins	I-paraffins	Olefins	Naphthenes	Aromatics	Total
C3						0.000
C4						0.000
C5	0.052					0.052
C6	0.648	0.279		1.081	0.261	2.269
C7	6.821	3.411		4.885	4.303	19.420
C8	16.231	15.413		8.178	7.267	47.089
C9	8.338	10.984	0.440	7.455	1.584	28.801
C10	0.152	2.206	0.011			2.369
C11						0.000
Total Unknowns						0.000
Total	**32.242**	**32.293**	**0.451**	**21.599**	**13.415**	**100.000**

LABORATORY REPORT NO. CA-0028-2024

DETAILED HYDROCARBON ANALYSIS

Sample Descriptions / Label : Qatar Marine Crude Oil-Qatar Marine Loading dated 25-02-2024

FRACTION - (150 - 200°C)

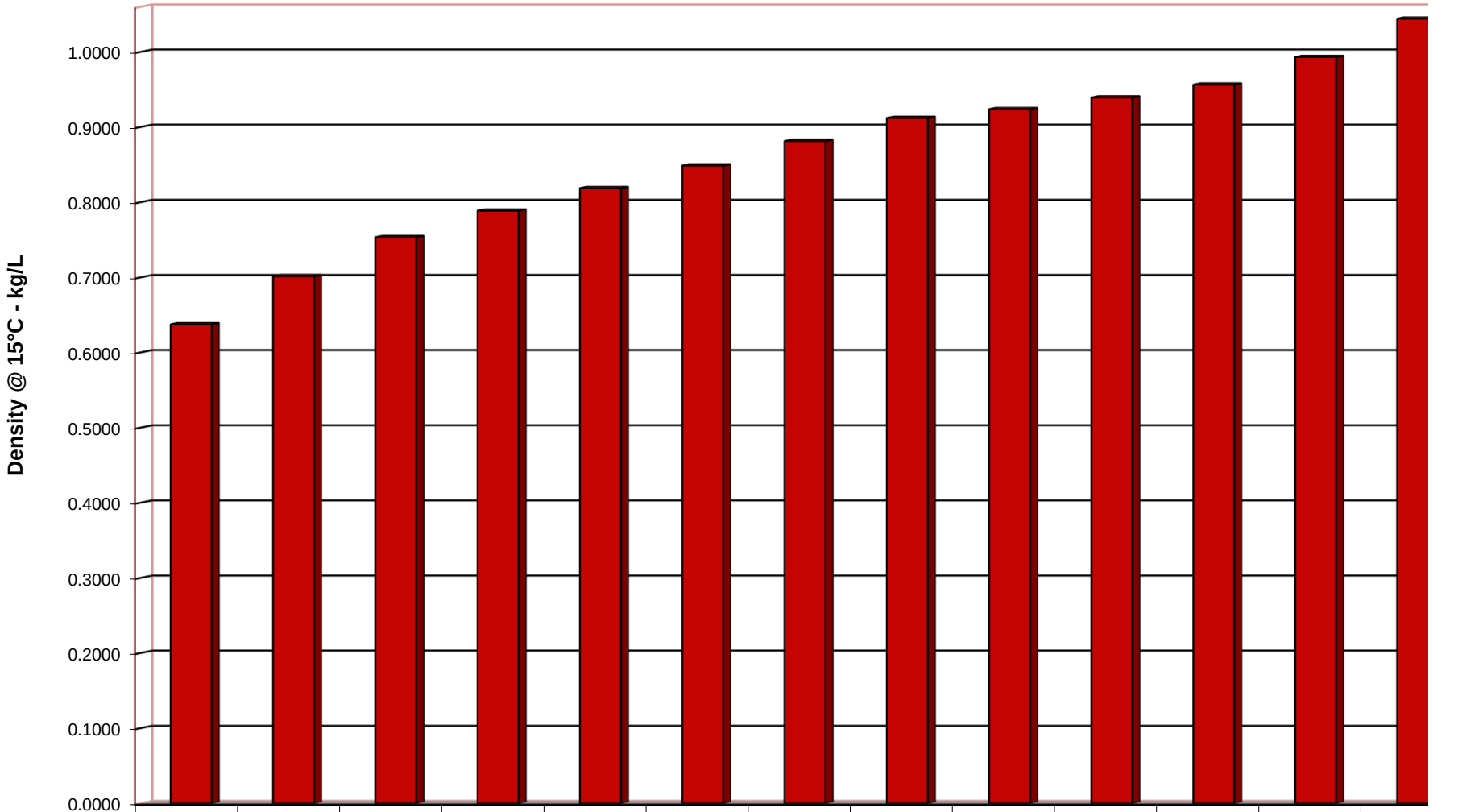
Totals by Group Type and Carbon Number (in Mass Percent)

	n-paraffins	I-paraffins	Olefins	Naphthenes	Aromatics	Total
C5						0.000
C6						0.000
C7					0.125	0.125
C8	0.503	0.091		0.703	2.203	3.500
C9	4.701	2.253	0.331	5.119	9.548	21.952
C10	9.176	10.505	2.014	13.784	7.733	43.212
C11	5.986	4.933	1.261	9.163	2.371	23.714
C12	1.372	4.231	0.170	1.316	0.408	7.497
C13						0.000
Total Unknowns						0.000
Total	**21.738**	**22.013**	**3.776**	**30.085**	**22.388**	**100.000**

Totals by Group Type and Carbon Number (in Volume Percent)

	n-paraffins	I-paraffins	Olefins	Naphthenes	Aromatics	Total
C5						0.000
C6						0.000
C7					0.116	0.116
C8	0.554	0.097		0.723	1.983	3.357
C9	5.133	2.459	0.341	5.111	8.588	21.632
C10	9.857	11.214	2.126	13.574	6.916	43.687
C11	6.344	5.077	1.314	8.833	2.124	23.692
C12	1.429	4.299	0.187	1.250	0.351	7.516
C13						0.000
Total Unknowns						0.000
Total	**23.317**	**23.146**	**3.968**	**29.491**	**20.078**	**100.000**

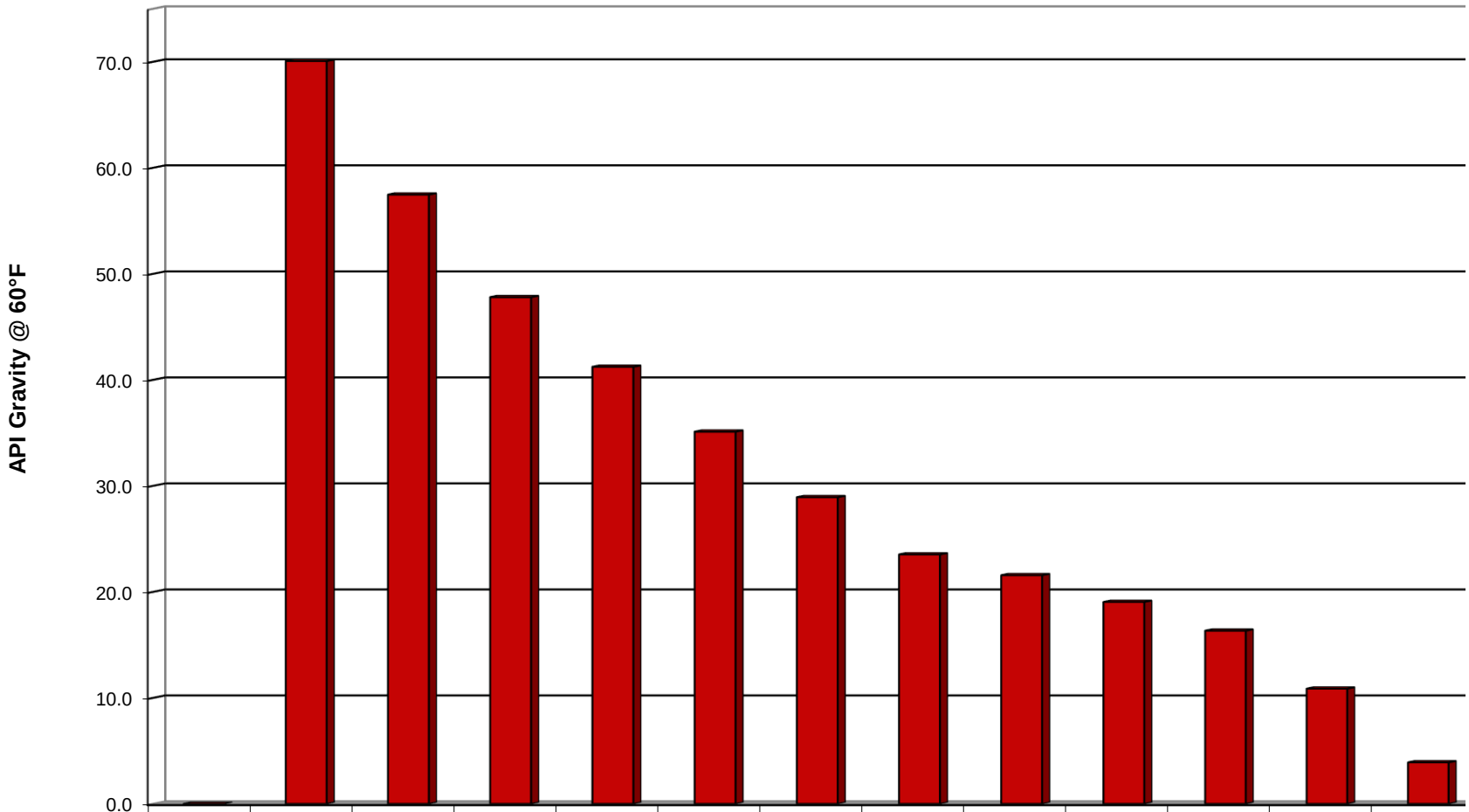
Fractions Mid B.P. Vs Density @ 15°C



	Naphtha (C5-50)	Naphtha (50-100)	Naphtha (100-150)	Kerosene (150-200)	Kerosene (200-250)	Gas Oil (250-300)	Gas Oil (300-350)	VGO (350-400)	VGO (400-450)	VGO (450-500)	VGO (500-550)	ATM RES (400 + Res)	Vacuum Res (550 + Res)
Density@15°C	0.6376	0.7018	0.7536	0.7888	0.8188	0.8489	0.8815	0.9122	0.9239	0.9394	0.9565	0.9933	1.0442

Fraction

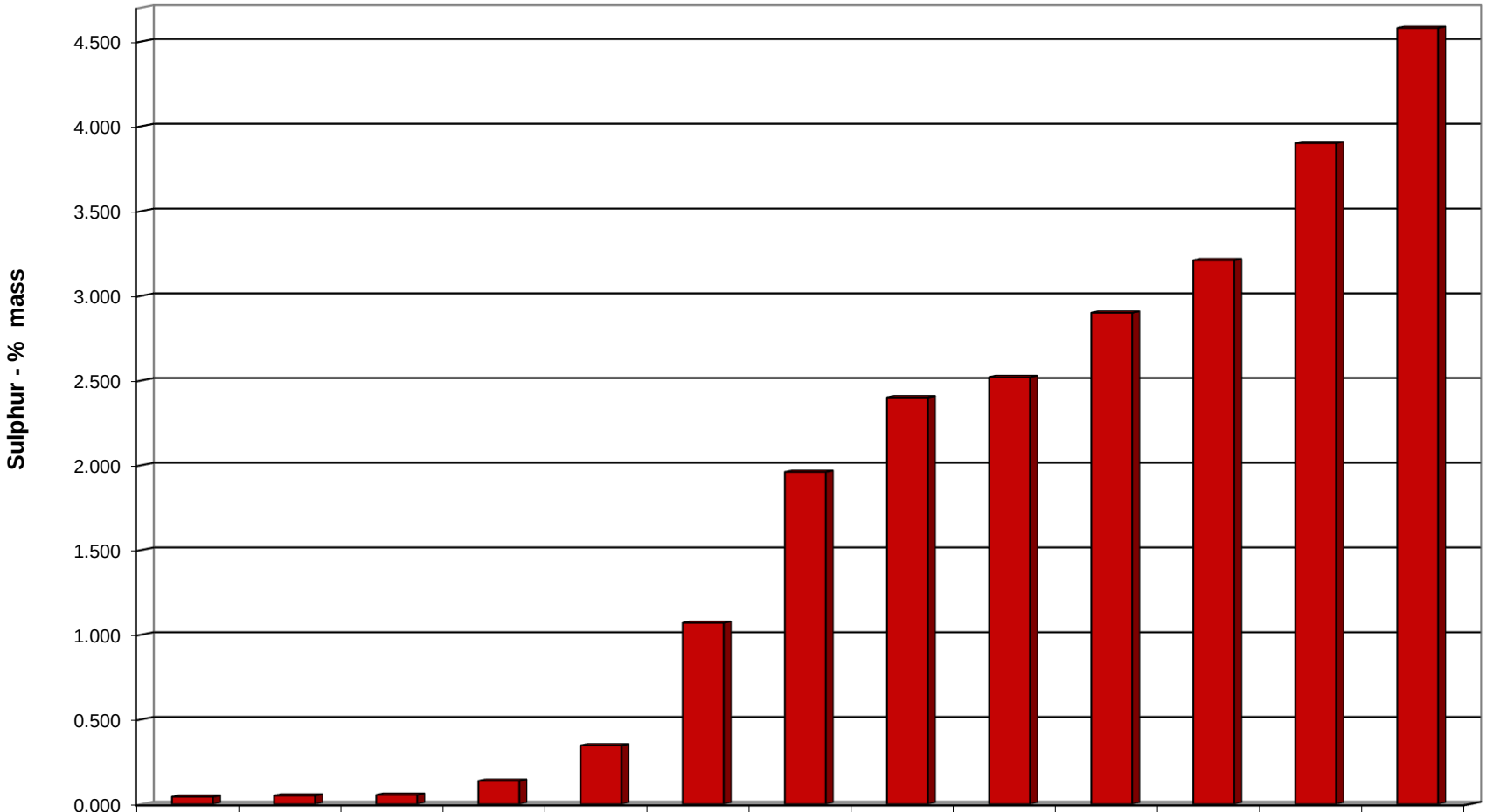
Fractions Mid B.P. Vs API Gravity @ 60°F



	Naphtha (C5-50)	Naphtha (50-100)	Naphtha (100-150)	Kerosene (150-200)	Kerosene (200-250)	Gas Oil (250-300)	Gas Oil (300-350)	VGO (350-400)	VGO (400-450)	VGO (450-500)	VGO (500-550)	ATM RES (400 + Res)	Vacuum Res (550 + Res)
API Gravity @60°F	0.0	70.1	57.5	47.8	41.2	35.1	28.9	23.5	21.6	19.1	16.3	10.9	3.9

Fraction

Fractions Mid B.P. Vs Sulphur



	Naphtha (C5-50)	Naphtha (50-100)	Naphtha (100-150)	Kerosene (150-200)	Kerosene (200-250)	Gas Oil (250-300)	Gas Oil (300-350)	VGO (350-400)	VGO (400-450)	VGO (450-500)	VGO (500-550)	ATM RES (400 + Res)	Vacuum Res (550 + Res)
Sulphur	0.0442	0.0510	0.0553	0.138	0.346	1.07	1.96	2.40	2.52	2.90	3.21	3.90	4.58

Fraction